Homework 3

BSTA 551

Directions

Same as previous instructions

For simulation problems, set your seed to `set.seed(551)` for reproducibility.

For the qmd of this Homework page, see this github link

You must show all of your work to receive credit.

This homework covers material from Lessons 6 and 7: Review of Point Estimation and Introduction to Confidence Intervals (pivotal quantities, two-sided z-intervals).

Questions

Problem 1: Pivotal Quantities and Confidence Intervals (Devore Section 8.1, Example 8.5 style)

A theoretical model suggests that the time-to-failure (in hours) of a certain medical device component follows an Exponential distribution with unknown rate parameter $\lambda > 0$. A random sample of $n = 8$ components yields the following failure times:

$$X = \{145, 89, 203, 67, 112, 156, 178, 94\}$$

Hint: If $X_1, \ldots, X_n \overset{iid}{\sim}$ Exponential(λ), then $2\lambda \sum_{i=1}^{n} X_i \sim \chi^2_{2n}$ (chi-squared with $2n$ degrees of freedom).

(a) Let $h(X_1, \ldots, X_n, \lambda) = 2\lambda \sum_{i=1}^{n} X_i$. Show that h is a pivotal quantity by verifying the definition included in class slides.

(b) Using the pivotal quantity from part (a), derive a $100(1-\alpha)\%$ confidence interval for λ. Express your answer in terms of $\sum X_i$ and chi-squared critical values $\chi^2_{\alpha/2,2n}$ and $\chi^2_{1-\alpha/2,2n}$.

(c) Compute a 95% confidence interval for λ using the data. In R, use `qchisq(p, df)` to find chi-squared quantiles.

(d) Find a 95% confidence interval for the mean time-to-failure $\mu = 1/\lambda$.

Problem 2: z-Interval for a Normal Mean (Devore Section 8.1)

A pharmaceutical company is conducting quality control on the fill volume (in mL) of syringes. Based on extensive historical data, the fill volumes are known to follow a normal distribution with standard deviation $\sigma = 0.05$ mL. The target fill volume is $\mu_0 = 1.00$ mL.

A random sample of $n = 20$ syringes from a new production batch yields a sample mean of $\bar{x} = 0.982$ mL.

(a) Verify that $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ is a pivotal quantity for μ (verify the definition in the slides, you can just state the distribution of $\bar{X}$ to start, don't need to derive it mathematically).

(b) Calculate a 95% confidence interval for the true mean fill volume μ using the observed sample mean provided.

(c) Based on your confidence interval, is there evidence that the true mean fill volume differs from the target of 1.00 mL? Explain.

(d) The company wants to estimate the mean fill volume to within ± 0.02 mL with 95% confidence. What sample size is required?

Problem 3: Devore Section 8.1, Exercise 1

Consider a normal population distribution with the value of σ known.

(a) What is the confidence level for the interval $\bar{x} \pm 2.81 \cdot \sigma/\sqrt{n}$? (Hint, use the definition of critical value and the `pnorm()` function in R – do sanity checks to make sure you're using the appropriate tail!).

(b) What is the confidence level for the interval $\bar{x} \pm 1.44 \cdot \sigma/\sqrt{n}$?

(c) What value of $z_{\alpha/2}$ in the CI formula results in a confidence level of 99.7%?

(d) Answer the question posed in part (c) for a confidence level of 75%.

Problem 4: Estimating Different Population Characteristics (Devore 7.4 Style)

A clinical researcher is studying verbal IQ scores in a population of patients recovering from mild traumatic brain injury. A random sample of $n = 12$ patients yields the following verbal IQ scores:

$$X = \{94, 108, 87, 115, 102, 99, 91, 106, 112, 98, 103, 89\}$$

Assume verbal IQ scores in this population follow a normal distribution with unknown mean μ and unknown variance σ^2.

(a) Compute point estimates for:

- The population mean μ
- The population variance σ^2 (use the unbiased estimator S^2)
- The population standard deviation σ

(b) The researcher wants to estimate the proportion of patients in this population with verbal IQ scores below 95 (the clinical threshold for concern). If $X \sim N(\mu, \sigma^2)$, then this proportion is $p = P(X < 95) = \Phi\left(\frac{95-\mu}{\sigma}\right)$.

Using the invariance property of MLEs and your estimates from part (a), compute a point estimate $\hat{p}$ for this proportion.

Hint: In R, use `pnorm()` to compute the standard normal CDF Φ.

(c) A colleague suggests a different estimator for p: simply use the sample proportion of observations below 95, i.e., $\tilde{p} = \frac{\#\{X_i < 95\}}{n}$.

- Compute $\tilde{p}$ for the given data
- Is $\tilde{p}$ unbiased for $p = P(X < 95)$? Explain briefly.
- (optional bonus question, do not need to complete for credit) Which estimator ($\hat{p}$ or $\tilde{p}$) do you think makes better use of the normality assumption? Why?

(d) Using R, compute the estimated standard error of $\bar{X}$:

$$\widehat{SE}(\bar{X}) = \frac{S}{\sqrt{n}}$$

If a different researcher collected a new sample of the same size, how much would you expect their sample mean to differ from yours, roughly speaking?

```r
# Data
iq_scores <- c(94, 108, 87, 115, 102, 99, 91, 106, 112, 98, 103, 89)

# Your code here
```